

i) Insert one document of nested levels [\[single_ecomm_doc.json\]](#)

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists several connections, including 'eCommerce' which is expanded to show 'eCommerceDB' and 'ecommerce'. The main editor on the right is titled '>_MONGOSH' and displays a JavaScript command to insert a document into the 'ecommerce' database. The document is a nested JSON object with the following structure:

```
> db.ecommerce.insertOne({
  "customerID": "CUST0026",
  "personalInfo": {
    "name": {
      "firstName": "Prady_26",
      "lastName": "Mohta_26"
    },
    "contact": {
      "email": "prady26@example.com",
      "phone": "261-555-3329",
      "address": {
        "street": "220 Guliford Road",
        "city": "Bethesda",
        "state": "NY",
        "zipCode": "38855",
        "country": "USA",
        "deliveryDetails": {
          "instructions": "Leave at front door",
          "timePreferences": {
            "weekdays": "Afternoon",
            "weekends": "Morning",
            "specialInstructions": {
              "ifNotHome": "Leave with neighbor",
              "neighborDetails": {
                "name": "Manav Gupta",
                "contactNumber": "333-444-5555",
                "address": "4321 Hartwick Road"
              }
            }
          }
        }
      }
    }
  },
  "viewedProducts": [
```

Compass

My Queries

CONNECTIONS (2)

Search connections

eCommerce

admin

config

eCommerceDB

ecommerce

local

organization

test

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>_MONGOSH

```
    }
  ]
},
{
  "category": "Books",
  "subCategory": "Subcategory 2",
  "items": [
    {
      "name": "Product 2",
      "brand": "Brand 2",
      "details": {
        "model": "Model 2",
        "specs": {
          "price": 817,
          "features": {
            "dimension": "5 inches",
            "weight": "4 kg",
            "extras": {
              "warranty": "1 year",
              "color": "White"
            }
          }
        }
      }
    }
  ]
},
{
  "totalPrice": 274
}
])
< {
  acknowledged: true,
  insertedId: ObjectId('67f1675b3ff1be35c94316db')
}
> db
< eCommerceDB
```

eCommerce > eCommerceDB > ecommerce

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Generate query

Explain

Reset

Find

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

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```
_id: ObjectId('67f055c73b894ba8cd02acb0')
customerID: "CUST0026"
personalInfo: Object
viewedProducts: Array (1)
wishlist: Array (4)
cart: Object
orderHistory: Array (2)
```

ii) Insert multiple documents of nested levels [[multiple_ecomm_doc.json](#)]

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists several databases, with 'eCommerceDB' expanded to show the 'ecommerce' collection. The main area displays a MongoDB shell prompt with a nested JSON document being inserted into the 'ecommerce' collection. The document contains customer information, contact details, and delivery preferences.

```
>_MONGOSH
> db.ecommerce.insertMany([
  {
    "customerID": "CUST0027",
    "personalInfo": {
      "name": {
        "firstName": "Prady_27",
        "lastName": "Mohta_27"
      },
    },
    "contact": {
      "email": "prady27@example.com",
      "phone": "680-555-4671",
      "address": {
        "street": "916 Guliford Road",
        "city": "Rock Ville",
        "state": "VA",
        "zipCode": "34790",
        "country": "USA",
        "deliveryDetails": {
          "instructions": "Leave at front door",
          "timePreferences": {
            "weekdays": "Afternoon",
            "weekends": "Morning",
            "specialInstructions": {
              "ifNotHome": "Leave with neighbor",
              "neighborDetails": {
                "name": "Manav Gupta",
                "contactNumber": "333-444-5555",
                "address": "4321 Hartwick Road"
              }
            }
          }
        }
      }
    }
  }
])
```

Compass

My Queries

CONNECTIONS (2)

Search connections

eCommerce

admin

config

eCommerceDB

ecommerce

local

organization

test

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eCommerce

ecommerce

mongosh: eCommerce

>_MONGOSH

```
}
]
},
{
  "category": "Books",
  "subCategory": "Subcategory 2",
  "items": [
    {
      "name": "Product 2",
      "brand": "Brand 2",
      "details": {
        "model": "Model 2",
        "specs": {
          "price": 817,
          "features": {
            "dimension": "5 inches",
            "weight": "4 kg",
            "extras": {
              "warranty": "1 year",
              "color": "White"
            }
          }
        }
      }
    }
  ]
},
],
"totalPrice": 274
}
]
})
< {
  acknowledged: true,
  insertedId: ObjectId('67f1675b3ff1be35c94316db')
}
> db
< eCommerceDB
```

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Generate query Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE 25 26 - 28 of 28

```
{
  "_id": ObjectId("67f055c73b894ba8cd02acb0"),
  "customerID": "CUST0026",
  "personalInfo": Object,
  "viewedProducts": Array (1),
  "wishlist": Array (4),
  "cart": Object,
  "orderHistory": Array (2)
}
```

```
{
  "_id": ObjectId("67f088913b894ba8cd02acb7"),
  "customerID": "CUST0027",
  "personalInfo": Object,
  "viewedProducts": Array (3),
  "wishlist": Array (3),
  "cart": Object,
  "orderHistory": Array (2)
}
```

```
{
  "_id": ObjectId("67f088913b894ba8cd02acb8"),
  "customerID": "CUST0028",
  "personalInfo": Object,
  "viewedProducts": Array (4),
  "wishlist": Array (2),
  "cart": Object,
  "orderHistory": Array (1)
}
```

Compass

My Queries

CONNECTIONS (2)

Search connections

eCommerce

- University
- admin
- config
- eCommerceDB
 - ecommerce
 - local
 - organization
 - test
- Robert Smith, MD

Welcome eCommerce ecommerce mongosh: eCommerce

```
>_MONGOSH
> db.eCommerce.find({ "customerID": { $in: ["CUST0026", "CUST0027", "CUST0028"] } }).pretty()
< {
  "_id": ObjectId('67f090d043efe989a40e5d78'),
  "customerID": 'CUST0026',
  "personalInfo": {
    "name": {
      "firstName": 'Prady_26',
      "lastName": 'Mohta_26'
    },
  },
  "contact": {
    "email": 'prady26@example.com',
    "phone": '261-555-3329',
    "address": {
      "street": '220 Guliford Road',
      "city": 'Bethesda',
      "state": 'NY',
      "zipCode": '38855',
      "country": 'USA',
      "deliveryDetails": {
        "instructions": 'Leave at front door',
        "timePreferences": {
          "weekdays": 'Afternoon',
          "weekends": 'Morning',
          "specialInstructions": {
            "ifNotHome": 'Leave with neighbor',
            "neighborDetails": {
```

iii) Delete a customer who has viewed products of Electronics category, has a wish list category of Books and has placed an order where the total price is greater than 1000.

```
> db.ecommerce.deleteOne({
  "viewedProducts.category": "Electronics",
  "wishlist.category": "Books",
  "orderHistory.totalPrice": { $gt: 1000 }
})
< {
  acknowledged: true,
  deletedCount: 1
}
```

iv) Delete customers living in MD and having order history of less than 1000.

```
> db.ecommerce.deleteMany({
  "personalInfo.contact.address.state": "MD",
  "orderHistory.totalPrice": { $lt: 1000 }
})
< {
  acknowledged: true,
  deletedCount: 7
}
```

v) If a customer has viewed products of category book with cart items having price greater than 100, change his email and set first item in his wish list priority of High.

```
> db.ecommerce.updateMany({
  "viewedProducts.category": "Books",
  "cart.items.items.details.specs.price": { $gt: 100 }
},{ $set: {
  "personalInfo.contact.email": "updated_to_new@example.com",
  "wishlist.0.priority": "High"
}
})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 8,
  upsertedCount: 0
}
```

vi) Find customers who have items in cart greater than \$100 and have viewed Electronic Products

```
> db.ecommerce.find({
  "cart.items.items.details.specs.price": { $gt: 100 },
  "viewedProducts.category": "Electronics"
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec72b'),
  customerID: 'CUST0007'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec731'),
  customerID: 'CUST0013'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec733'),
  customerID: 'CUST0015'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec739'),
  customerID: 'CUST0021'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec73c'),
  customerID: 'CUST0024'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec73d'),
  customerID: 'CUST0025'
}
eCommerceDB >
```


>_ mongosh: myConnection



>_MONGOSH

```
> db.ecommerce.find({
  "cart.items.items.details.specs.price": { $gt: 100 },
  "viewedProducts.category": "Electronics"
})
```

```
< {
  _id: ObjectId('67f1637d1267e1e5b12ec72b'),
  customerID: 'CUST0007',
  personalInfo: {
    name: {
      firstName: 'Prady_7',
      lastName: 'Mohta_7'
    },
    contact: {
      email: 'prady7@example.com',
      phone: '731-555-5317',
      address: {
        street: '284 Guliford Road',
        city: 'Bethesda',
        state: 'VA',
        zipCode: '61151',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
              neighborDetails: {
```

vii) Find customers who are from MD and have viewed products of category both Electronics and Books

```
> db.ecommerce.find({
  "personalInfo.contact.address.state": "MD",
  "viewedProducts.category": { $all: ["Electronics", "Books"] }
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec733'),
  customerID: 'CUST0015'
}
```

>_ mongosh: myConnection



>_MONGOSH

```
> db.ecommerce.find({
  "personalInfo.contact.address.state": "MD",
  "viewedProducts.category": {$all: ["Electronics", "Books"]}
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec733'),
  customerID: 'CUST0015',
  personalInfo: {
    name: {
      firstName: 'Prady_15',
      lastName: 'Mohta_15'
    },
    contact: {
      email: 'updated_to_new@example.com',
      phone: '287-555-4363',
      address: {
        street: '885 Guliford Road',
        city: 'Adelphi',
        state: 'MD',
        zipCode: '20507',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
              neighborDetails: {
```

viii) Find customers who have products of Footwear in their wish list and have viewed Home Appliances

```
> db.ecommerce.find({
  "wishlist.category": "Footwear",
  "viewedProducts.category": "Home Appliances"
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec725'),
  customerID: 'CUST0001'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec729'),
  customerID: 'CUST0005'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec72c'),
  customerID: 'CUST0008'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec735'),
  customerID: 'CUST0017'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec73a'),
  customerID: 'CUST0022'
}
eCommerceDB >
```

>_ mongosh: myConnection



>_MONGOSH

>

```
db.ecommerce.find({
  "wishlist.category": "Footwear",
  "viewedProducts.category": "Home Appliances"
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec725'),
  customerID: 'CUST0001',
  personalInfo: {
    name: {
      firstName: 'Prady_1',
      lastName: 'Mohta_1'
    },
    contact: {
      email: 'prady1@example.com',
      phone: '810-555-5903',
      address: {
        street: '968 Guliford Road',
        city: 'Bethesda',
        state: 'VA',
        zipCode: '49836',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
```

ix) Find Customers who have ordered more than \$1500 goods and live in College Park.

```
> db.ecommerce.find({
  "orderHistory.totalPrice": {$gt : 1500},
  "personalInfo.contact.address.city": "College Park"
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec734'),
  customerID: 'CUST0016'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec738'),
  customerID: 'CUST0020'
}
```

>_ mongosh: myConnection



>_MONGOSH

```
> db.ecommerce.find({
  "orderHistory.totalPrice": {$gt : 1500},
  "personalInfo.contact.address.city":"College Park"
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec734'),
  customerID: 'CUST0016',
  personalInfo: {
    name: {
      firstName: 'Prady_16',
      lastName: 'Mohta_16'
    },
    contact: {
      email: 'prady16@example.com',
      phone: '599-555-7765',
      address: {
        street: '510 Guliford Road',
        city: 'College Park',
        state: 'VA',
        zipCode: '24412',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
              neighborDetails: {
```

x) Find customers who have viewed Electronic Products and have items which are of Electronics category in their cart.

```
> db.ecommerce.find({
  "viewedProducts.category": "Electronics",
  "cart.items.category": "Electronics"
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec73c'),
  customerID: 'CUST0024'
}
```


>_ mongosh: myConnection



>_MONGOSH

```
> db.ecommerce.find({
  "viewedProducts.category": "Electronics",
  "cart.items.category": "Electronics"
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec73c'),
  customerID: 'CUST0024',
  personalInfo: {
    name: {
      firstName: 'Prady_24',
      lastName: 'Mohta_24'
    },
    contact: {
      email: 'prady24@example.com',
      phone: '829-555-4892',
      address: {
        street: '812 Guliford Road',
        city: 'College Park',
        state: 'VA',
        zipCode: '10708',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
              neighborDetails: {
```

xi) Find customers who have viewed Clothing products, have Books in their wishlist and live in Hyattsville.

```
> db.ecommerce.find({
  "viewedProducts.category": "Clothing",
  "wishlist.category": "Books",
  "personalInfo.contact.address.city": "Hyattsville"
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec731'),
  customerID: 'CUST0013'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec739'),
  customerID: 'CUST0021'
}
```

>_ mongosh: myConnection



>_MONGOSH

```
> db.ecommerce.find({
  "viewedProducts.category": "Clothing",
  "wishlist.category": "Books",
  "personalInfo.contact.address.city": "Hyattsville"
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec731'),
  customerID: 'CUST0013',
  personalInfo: {
    name: {
      firstName: 'Prady_13',
      lastName: 'Mohta_13'
    },
    contact: {
      email: 'prady13@example.com',
      phone: '507-555-4005',
      address: {
        street: '366 Guliford Road',
        city: 'Hyattsville',
        state: 'MD',
        zipCode: '83986',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
```

xii) Find customers who have placed an order in the year 2023, their cart items have a warranty and cart items have a category of Home Appliances.

```
> db.ecommerce.find({
  "orderHistory.orderDate":{$regex:"^2023"},
  "orderHistory.products.items.details.specs.features.extras.warranty":{$exists:true},
  "cart.items.category":"Home Appliances"
}, {customerID: 1})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec725'),
  customerID: 'CUST0001'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec729'),
  customerID: 'CUST0005'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec72b'),
  customerID: 'CUST0007'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec72d'),
  customerID: 'CUST0009'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec730'),
  customerID: 'CUST0012'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec737'),
  customerID: 'CUST0019'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec739'),
  customerID: 'CUST0021'
}
{
  _id: ObjectId('67f1637d1267e1e5b12ec73d'),
  customerID: 'CUST0025'
}
```

```
{
  _id: ObjectId('67f167c03ff1be35c94316dc'),
  customerID: 'CUST0027'
}
```

mongosh: myConnection



> _MONGOSH

```
> db.ecommerce.find({
  "orderHistory.orderDate":{$regex:"^2023"},
  "orderHistory.products.items.details.specs.features.extras.warranty":{$exists:true},
  "cart.items.category":"Home Appliances"
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec725'),
  customerID: 'CUST0001',
  personalInfo: {
    name: {
      firstName: 'Prady_1',
      lastName: 'Mohta_1'
    },
    contact: {
      email: 'prady1@example.com',
      phone: '810-555-5903',
      address: {
        street: '968 Guliford Road',
        city: 'Bethesda',
        state: 'VA',
        zipCode: '49836',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
            }
          }
        }
      }
    }
  }
}
```

xiii) Find customers who live in VA, have viewed products of Electronics and have a product less than \$200 in their cart.

```
> db.ecommerce.find({
  "personalInfo.contact.address.state": "VA",
  "viewedProducts.category": "Electronics",
  "cart.items.items.details.specs.price": { $lt : 200 }
}, { customerID: 1 })
< {
  _id: ObjectId('67f1637d1267e1e5b12ec72b'),
  customerID: 'CUST0007'
}
```

>_ mongosh: myConnection



>_MONGOSH

```
> db.ecommerce.find({
  "personalInfo.contact.address.state": "VA",
  "viewedProducts.category": "Electronics",
  "cart.items.items.details.specs.price": { $lt : 200 }
})
< {
  _id: ObjectId('67f1637d1267e1e5b12ec72b'),
  customerID: 'CUST0007',
  personalInfo: {
    name: {
      firstName: 'Prady_7',
      lastName: 'Mohta_7'
    },
    contact: {
      email: 'prady7@example.com',
      phone: '731-555-5317',
      address: {
        street: '284 Guilford Road',
        city: 'Bethesda',
        state: 'VA',
        zipCode: '61151',
        country: 'USA',
        deliveryDetails: {
          instructions: 'Leave at front door',
          timePreferences: {
            weekdays: 'Afternoon',
            weekends: 'Morning',
            specialInstructions: {
              ifNotHome: 'Leave with neighbor',
```